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Study on the influence mechanism of different grassland utilization on soil preferential flow in semi-arid grassland in northern China

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ABSTRACT

The semi-arid grasslands of northwestern China are characterized by fragile ecosystems, limited water resources, and significant degradation. This study investigates the effects of different land-use practices, such as fencing, grazing, and tourism, on soil preferential flow in the semi-arid grasslands of central Inner Mongolia. The preferential flow process was observed using the dye tracer method and quantitatively analyzed via image processing to determine morphological features. Results indicated that preferential flow was observed in all areas, but the extent varied significantly: fenced areas exhibited the most pronounced preferential flow at both 25 mm and 50 mm infiltration volumes. Preferential flow mainly occurred in the top 30 cm, with a decreasing ratio of dyed area at greater depths. At 25 mm infiltration, the preferential flow ratio exceeded 50% in fenced areas. Increasing the volume to 50 mm resulted in peak values for infiltration depth and stained area ratio, although the proportion of preferential flow decreased. Key factors influencing preferential flow included soil bulk density, texture, and root nutrient characteristics. This study provides essential insights for effective grassland management and water resource conservation, contributing to the governance of semi-arid ecosystems by prioritizing fencing to improve soil health.

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Introduction

Grasslands are Earth's most extensive terrestrial ecosystem, covering approximately 40% of the planet's land surface and contributing to 30% of the global terrestrial carbon (Lin et al. 2022). The unsustainable use of grassland resources and climate change, however, have made grassland degradation a significant worldwide ecological concern (Chen and Tang 2016). Long-term overgrazing and excessive tourism have severely degraded the soil and vegetation in Inner Mongolia's semi-arid grasslands, which are characterized by delicate climatic conditions and are at risk of desertification. Since its spatial distribution and dynamics directly affect ecosystem biodiversity, soil moisture serves as the main constraint on vegetation growth and recovery in semi-arid regions (Zhu et al. 2020). According to Jimoh et al. (2020), maintaining soil water storage is crucial for the ecological restoration of semi-arid grasslands. In these water-limited environments, understanding the hydrological processes that govern water infiltration and redistribution is paramount, as they are highly sensitive to land use and climate variability (Zhao et al. 2023). Water bypasses the soil matrix through wormholes, macropores, fractures, and plant roots, allowing for preferential flow and quick vertical movement. In addition to surpassing matrix flow velocities, this gravity-driven process functions independently of capillary forces (Nimmo 2020). They have preferential flow.

Current research on preferential flow focuses on its developmental patterns and regulatory mechanisms, which are key yet challenging topics that have been explored in ecological and hydrological studies. Factors affecting soil preferential flow include vegetation, biological activity, topography, soil type and structure, and hydraulic properties, which collectively provide preferential pathways (Lin 2010; Wiekenkamp et al. 2016). Soils exhibit distinct textures, structures, water retention capacities, and pore networks, creating hydrological conditions conducive to preferential flow. Uncertainty in these processes is particularly high in semi-arid regions, necessitating robust characterization methods (Zhao et al. 2023). Critical macropore characteristics

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such as aperture size, curvature, and connectivity directly impact preferential flow dynamics (Budhathoki et al. 2022). Both living and dead plant roots create preferential pathways that improve water infiltration, especially in arid areas (Zhang et al. 2022). The establishment of preferred flow is concurrently regulated by pore networks, rainfall intensity, and duration (Zhang et al. 2018). While mild rainfall merely moistens topsoil without replenishing groundwater, intense rainfall episodes cause water to quickly enter fractures, penetrate capillary barriers, and reach deeper soil layers (Galeandro et al. 2014). The impact of such precipitation patterns on ecological processes is profound, extending to nutrient cycling in arid systems (Pang et al. 2024). Together, the previously described factors affect the occurrence of soil preferential flow. The extent and direction of their influence, however, vary widely and even run counter to the findings of other researchers. This suggests that each factor's effect is contingent upon the research area's geographic setting.

The effects of varying grazing intensities and enclosure treatments on vegetation and soil physicochemical properties are the primary focus of semi-arid grasslands research (Chen and Tang 2016; Jimoh et al. 2020). Recent studies have advanced the monitoring of ecological security patterns based on land use changes (Zhang J et al. 2025) and the delineation of groundwater resources using geospatial techniques (Nazir et al. 2024), highlighting the critical role of hydrological processes in ecosystem health. However, few studies have investigated how grazing, enclosure, and tourism influence soil preferential flow. This makes it difficult to comprehensively understand how different land uses alter soil and root characteristics and how these changes further influence soil preferential flow processes. Additionally, precipitation events are scarce and highly fluctuating in arid and semi-arid regions (Galeandro et al. 2014), a fact supported by global analyses of extreme sub-daily precipitation patterns (Zhang Y and Wu X 2025), which makes it essential to depict the variations in soil preferential flow characteristics after diverse rainfall events (Wu et al. 2022). Current literature inadequately describes or quantitatively tests preferential flow properties under varying precipitation conditions. It remains unclear whether infiltration impacts on preferential flow are consistent across grazed, fenced, and tourist grasslands, or whether preferential flow initiation and development differ among land-use types under identical infiltration conditions. Thus, further research is urgently needed, particularly regarding the effects of major land uses like grazing and tourism on preferential flow in these ecosystems.

This study aims to address the following objectives: (1) evaluate the impact of the methods of utilizing different grassland on soil properties and vegetation characteristics; (2) apply the dye tracer method to determine the characteristics of flow under the following infiltration conditions: 10 mm, 25 mm, and 50 mm; (3) identify the key factors affecting the preferential flow at the small scale in semi-arid grasslands.

This study also aims to fill the gap regarding the impact of different utilization methods on the preferential flow characteristics of semi-arid grasslands. Additionally, it determines quantitatively how various levels of infiltration affect soil preferred flow in fenced and grasslands available for tourists and grazing. It also attempts to explore the causes underlying the differences. Furthermore, policy recommendations to relevant authorities on how to manage semi-arid grasslands more effectively and sustainably have been provided.

Materials and methods

Study area

The site selected for the study is located in the southern part of Xilamuren Grassland, Xilamuren Town, Baotou City, Inner Mongolia (41°20'55"–41°21'23"N, 111°12'19"–111°12'43"E) (Figure 1). The area has mid-temperate semi-arid continental monsoon climate a semi-arid continental monsoon climate, with a mean annual temperature of 2.5°C and an average annual precipitation of 286.7 mm. The region exhibits four distinct seasons, abundant sunshine, and a considerable diurnal temperature variation. Spring is characterized by frequent winds, with an average of 26 windy days annually featuring persistently high wind speeds and intensity. Summers are hot with concentrated rainfall, primarily manifested as short-duration, high-intensity showers. Winters are cold and arid, frequently influenced by cold-air intrusions that result in cold waves, strong winds, and snowfall events. The dominant soils are chestnut calcium soil and meadow soil (IARRP et al. 2009). Grassland is the dominant land use, with vegetation predominantly composed of meso-xerophytic and xero-mesophytic plant species. The average vegetation coverage reaches 28%. *Stipa krylovii* Roshev, *Leymus chinensis*, *Agropyron cristatum*, *A. frigida* Willd, and *S. breviflora* Griseb are the dominant plant species (Huang et al. 2016; Xu et al. 2022). Xilamuren Town boasts abundant grassland resources, with utilizable

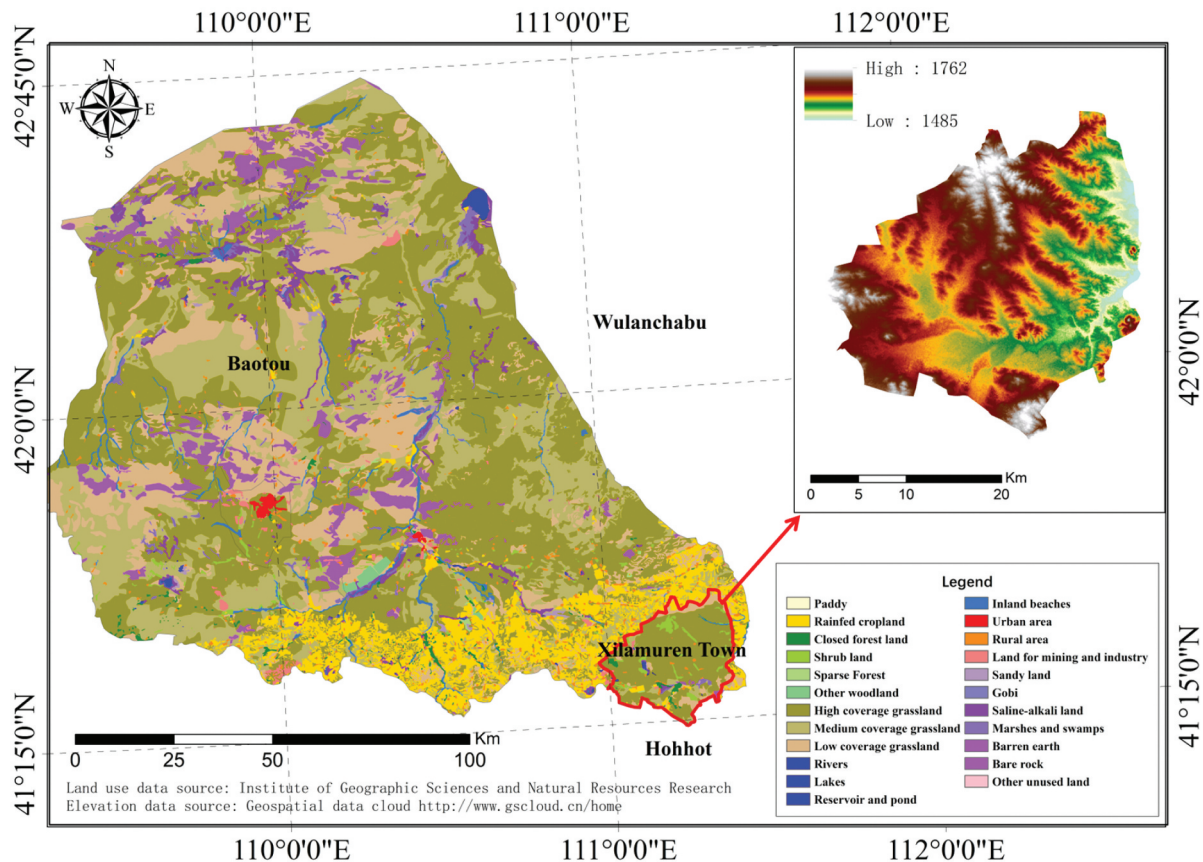


Figure 1. Geographical location of the study area.

pasture area reaching 690 km² and a grassland utilization rate exceeding 95%. The local economy is primarily driven by grassland tourism services and traditional animal husbandry. The Xilamuren grassland exhibits distinct characteristics of combined hydraulic and wind erosion, resulting in severe soil erosion. Coupled with long-term unsustainable practices such as overgrazing and tourist trampling, extensive areas of grassland have undergone severe degradation and desertification.

Field experimental design

Sample site selection

In the Xilamuren grassland, the three common modes of utilizing the grasslands are tourism, grazing, and fencing and sealing. Three areas of research were selected to represent modes of utilization: the grazing exclusion area (GE), the free grazing area (FG), and the tourism area (TA). All three regions are located nearby and have similar terrain and climate conditions. Hence, the three areas effectively eliminate the interference of non-management factors such as soil parent material, terrain, altitude, and precipitation. The GE area was enclosed with a 2.1 m metal fence in 2002, which completely prevented any disturbances due to grazing and tourism. The sample plot size is 150 m × 200 m. For more than 20 years, the FG region, which is east of GE, has been grazed at intensities of 1.5–2.0 sheep/ha-year (May – October) and 0.5–1.0 sheep/ha-year (November – April); the sample plot size is 165 m × 110 m. The northern parts of the GE and FG areas are where the tourist area is located. The region was transformed into a tourist destination in 2004. The peak tourist season is June to October, with a daily maximum of 4860 visitors. The sample plot is 125 m × 100 m.

Vegetation survey and soil sampling

Plant and soil samples were collected from the three sites between August 27 and 30 in the year 2021. The area did not receive any precipitation before or during sampling. Within each study region, three 20 m × 20 m

plots were randomly selected, and five 1 m × 1 m sample squares were randomly distributed per plot. Adjacent squares were separated by at least five meters. The species name, density, height, and cover (visually judged) of every plant within the squares were noted. In order to measure dry biomass, all plants in the sample squares were cut flush with the roots, weighed immediately for fresh biomass, and then oven-dried at 75°C until their weight remained constant. Soil samples were collected from the 0–40 cm soil layer from the four corners of each sample square. Soil samples were collected using four 100 cm³ soil loop knives.

Vegetation and soil measurement

In this study, four plant diversity indices were selected to measure and analyze the species diversity characteristics of the communities in the sample plots (Wang et al. 2018): Margalef richness index (R), Simpson dominance index (D), Shannon-Wiener diversity index (H'), and Pielou evenness index (E).

Margalef richness index (R):

$$R = (S - 1) / \ln N$$

Shannon-Wiener diversity index (H'):

$$H' = - \sum_{i=1}^S P_i \ln P_i$$

Simpson dominance index (D):

$$D = 1 - \sum_i^P 2$$

Pielou evenness index (E):

$$E = H' / \ln S$$

S represents the total number of species in the community; N represents the total number of individuals across all species in the community; and N_i represents the number of individuals of species i .

The soil was characterized and examined using six indicators. In particular, measurements of the soil's bulk density, water content, total porosity, and saturated hydraulic conductivity were made. Furthermore, four 100 cm³ soil samples were taken, three replicates per soil layer, in order to ascertain the mechanical composition and organic carbon content of the soil.

The potassium dichromate volumetric-external heating method was used to measure the amount of organic carbon in the soil (Pulido et al. 2017). A Mastersizer 3000 laser particle size analyzer was used to measure the particle size distribution of the soil. The water content of the soil was determined by oven-drying at 105°C for 24 hours until its weight remained constant (W_d). The ring-knife immersion method was used to measure the porosity of the soil. Additionally, the given equations were used to determine BD and TP.

$$BD = \frac{W_d - W_c}{V}$$

$$TP = \left(1 - \frac{BD}{ds} \right) \times 100$$

where V is the volume of the soil ring cutter (cm⁻³); W_c is the weight of the ring cutter (g); and ds is the soil particle density (2.65 g cm⁻³) in this study.

Soil dye tracer experiments

The preferential flow process was observed using the dye tracer method. This method is widely used in grassland preferential flow studies as a simple, low-cost approach (Jimoh et al. 2020). Dye tracers provide precise spatial data on preferential flow paths, including their number, depth, and continuity during infiltration processes (Li et al. 2020). In late August, the underground biomass of dominant plants such as *Carex kobomugi* and *Leymus chinensis* reached its peak levels (Wang et al. 2017). After reviewing the local weather forecast, it was decided to conduct a dye tracing experiment on 28 August 2021, alongside plant and soil sampling, as there was no rainfall in the area for five days

before and after the experiment. A total of 27 experiments (3 plots $\times$ 3 squares $\times$ 3 titers) were performed. Three of the five sample squares per plot were randomly selected for infiltration experiments with 10 mm, 25 mm, and 50 mm solutions (representing moderate, heavy, and torrential rainfall, respectively). Frames built of homemade aluminum alloy measuring (50 cm $\times$ 50 cm $\times$ 30 cm; plate thickness = 0.5 cm) were inserted 15 cm into the soil. 30 mm/h was used to infuse Brilliant Blue solution (4 g L⁻¹) (2.5 L, 6.25 L, and 12.5 L for 10 mm, 25 mm, and 50 mm rainfall, respectively) (Li et al. 2020). For 24 hours, a black plastic sheet was placed over the frames to keep out rain and evaporation. Following removal, a 40 cm $\times$ 40 cm core area (5 cm buffer from the frame) was used to excavate vertical soil profiles. The staining penetration was used to assess the depth of excavation. A Nikon D7000 camera (21 mm focal length, 30 cm lens-to-profile distance) was used to take pictures of each layer. A grayscale color card and a scale ruler were also used. Geometric distortion correction and light correction were performed on the field photographs using Photoshop software, based on the rectangular frame formed by the ruler placed at the time of photographing and the grayscale colorimetric card. The images were converted to binary images consisting of only 0 and 255 gray values by image thresholding (Flury et al. 1994). Using ImageJ software, each profile photo was processed into a binary matrix of 400 pixels by 400 pixels.

Following the excavation of the vertical profile, plant roots were gathered layer by layer. Bags of roots from the same depth were taken to the laboratory for examination. After rinsing off any remaining soil, they were allowed to air-dry organically. The samples were scanned using a root scanner (EPSON 1000XL), and root system parameters such as root length, diameter, area, volume, and so on were measured using the WinRHIZO software. After analysis, the roots were dried to a constant weight in paper bags at 65°C to determine the root biomass (BGB). Root mean diameter (mm), root length density (RLD, cm cm⁻³), root surface area density (RSAD, cm² cm⁻³), root mass density (RMD, mg cm⁻³), and root volume density (RMD, mg cm⁻³) were used to determine the properties of the roots. The definitions of root mass density (RMD, mg·cm⁻³) and root volume density (RVD, cm³·cm⁻³) were based on Hao et al. (2020).

Measurement of the staining characteristics of preferential flow

The binary matrix data imported into Excel were used to determine the morphological parameters of the stained images from horizontal and vertical soil profiles. Each parameter was calculated as follows:

Maximum staining depth. Maximum staining depth (*MaxD*) represents the maximum penetration depth achieved by the staining solution in the profile.

Dyed area ratio. Dyed area ratio (*DC*) refers to the percentage of stained patch area relative to the total area of the soil profile at a specific depth of the layer of soil (Flury et al. 1994):

$$DC = \left(\frac{D}{D + ND} \right) \times 100\%$$

where, *DC* is the stained area ratio; *D* and *ND* represent the stained and unstained area, respectively.

Depth of matrix flow. The depth of substrate flow (*UniFr*) is the critical soil depth that demarcates the boundary between the matrix flow zone and the preferential flow zone in a soil staining profile, and in this study, the soil depth when the stained area is $\geq 80\%$ on the vertical profile was taken as the depth of the substrate flow (van Schaik 2009). At a certain amount of infiltration water, the greater the depth of substrate flow, the more lagging the occurrence of soil preferential flow (van Schaik 2009).

Priority flow fraction. Priority flow fraction (*PF_{fr}*) is the ratio of the area of preferential flow zones to the overall stained area in the soil staining profile, when the priority flow fraction is more than 50%, it indicates that the priority flow effect is stronger than the substrate flow effect (Kan et al. 2019).

$$PF_{fr} = 1 - \frac{UniFr * W}{TotStAr}$$

where PF_{fr} is the stained area ratio of preferential flow zones in the soil profile (%); $UniFr$ is the corresponding soil profile depth of the matrix flow zone in the soil profile (cm); W is the horizontal stained width of the soil profile (cm); and $TotStAr$ is the total stained area of the soil profile (cm²).

Length index. When a soil profile is uniformly separated into many layers of a specified width, the length index (L_i) is the sum of the absolute differences between the lower and upper layers. The length index increases with the degree of preferential flow development (Bargués-Tobella et al. 2014).

The better the development of preferential flow, the greater the length index (Bargués-Tobella et al. 2014).

$$L_i = \sum_{i=1}^n |DC_{i+1} - DC_i|$$

where L_i is the sum of the absolute differences between the $i + 1$ th and i th layers of the soil profile (in%); n is the number of vertical soil layers; and DC_{i+1} and DC_i are the stained area ratios (%) for layers $i + 1$ and i , respectively.

Results

Soil, vegetation, and root characteristics

As shown in Table 1, the bulk density (BD) of the fenced area (GE) was significantly lower than the free-grazing area (FG) and the tourist area (TA) ($p < 0.05$). The soil water content (SWC) values at the three sampling locations varied significantly, with GE having a considerably higher SWC value than FG and TA. While there was no significant difference in the TP between FG and GE or FG and TA ($p > 0.05$), the TP of GE was significantly greater than that of TA ($p < 0.05$). While there was no significant difference in SOC between FG and TA ($p > 0.05$), GE's SOC was significantly greater than both FG's and TA's ($p < 0.05$). The three sampling sites' average soil clay particle contents varied significantly, with the GE sample site having the greatest average content (5.30%), followed by the FG sample site (4.56%) and the TA sample site (2.15%). GE > FG > TA was the order of powder grain content ($p < 0.05$). In contrast, the trend for sand content was TA > FG > GE ($p < 0.05$).

Root traits differed across the three sample sites (Figure 2). The root diameter, RMD, and RSAD exhibited a decreasing trend with increasing soil depth. The root diameter at GE was substantially larger than at FG and TA ($p < 0.05$), although there was no significant difference between FG and TA ($p > 0.05$). Root length density (RLD) in the 0–20 cm soil layer was GE > FG > TA, with GE showing the highest developed root systems. The RSAD values of GE and FG were considerably greater than TA at 0–30 cm soil depth ($p < 0.05$), and there was no discernible variation in RSAD amongst the studied sites at 30–40 cm soil depth ($p > 0.05$). RMD gradually reduced as soil depth increased, reaching its maximum in the topsoil layer (0–10 cm) and its minimum in the subsoil layer (30–40 cm). In every depth, RMD displayed GE > FG > TA.

There were significant differences in the vegetation characteristics among the three sample sites (Table 2), exceeding FG by 5.89 cm and 14.23% and TA by 11.88 cm and 21.54%. The Margalef richness index (R) of GE was significantly higher than that of FG and TA. The Simpson index (D) and Shannon-Wiener index (H') of FG

Table 1. Soil physiochemical properties of three sites.

Site	BD (g cm ⁻³)	SWC (%)	TP (%)	SOC (g kg ⁻¹)	Soil texture		
					Sand(%)	Silt(%)	Clay (%)
GE	1.42±	4.93±	46.76±	34.84±	74.27±	20.40±	5.30±
	0.10b	0.51a	2.65a	6.80a	3.50c	2.76a	0.76a
FG	1.62±	3.44±	40.50±	30.70±	78.53±	16.88±	4.56±
	0.19a	1.47b	2.20ab	7.50b	4.91b	3.53b	0.88ab
TA	1.68±	1.59±	38.75±	30.65±	86.90±	10.96±	2.15±
	0.02a	0.48c	2.50b	1.90b	1.31a	1.58c	0.27b

Values are means ± SD ($n = 9$). Different letters denote significant differences ($p < 0.05$) among different land use patterns.

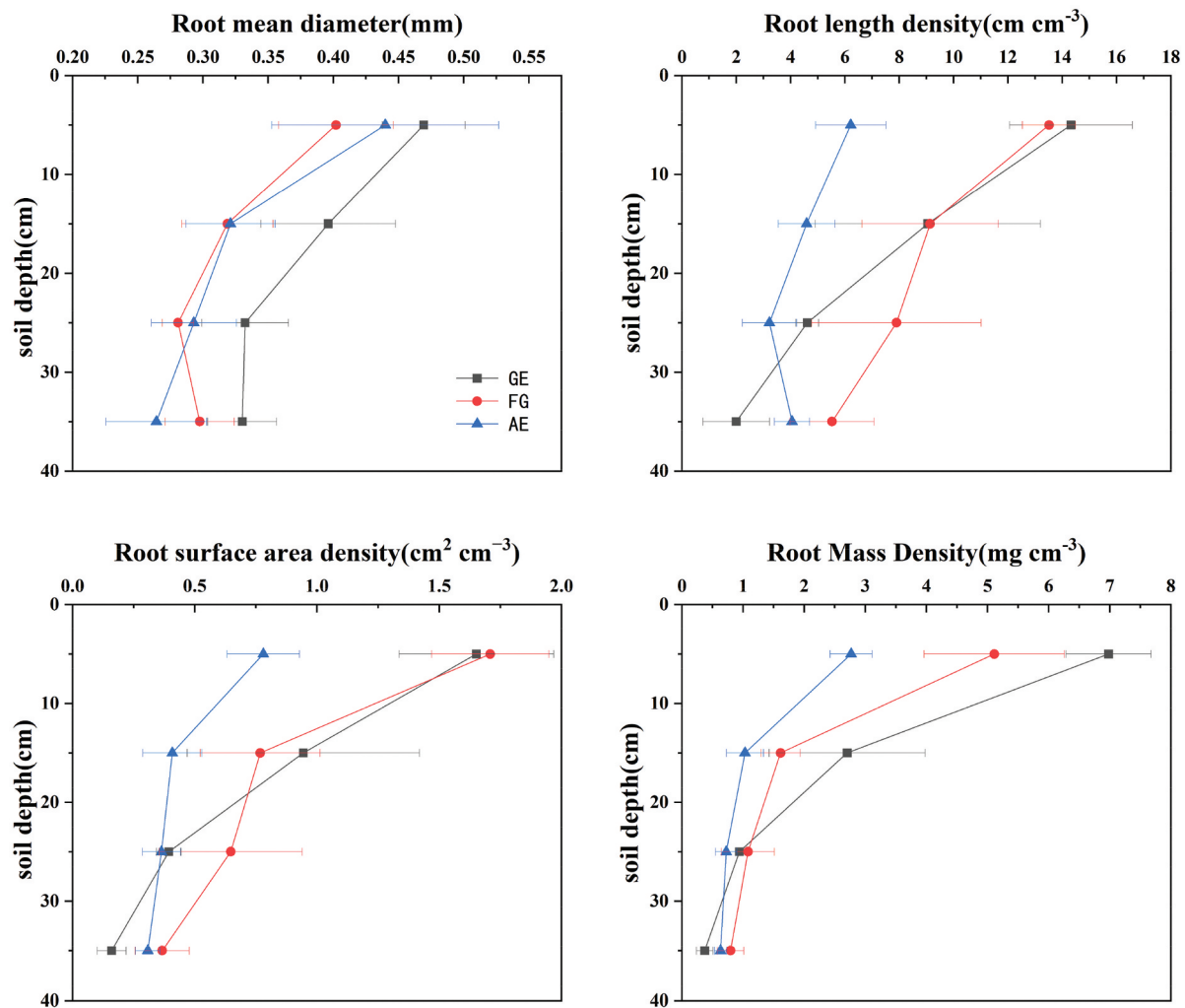


Figure 2. Root mean diameter, root length density, root surface area density and root mass density under different land use patterns of grassland.

Table 2. Geographic location and community characteristics under different land use patterns.

Site	Dominant plant species	Plant height (cm)	Vegetation coverage (%)	Margalef richness index (R)	Simpson index (D)	Shannon-Wiener index (H')	Pielou evenness (E)
GE	<i>Stipa krylovii</i> , <i>Leymus chinensis</i> , <i>Convolvulus ammannii</i> Desr	21.22±0.69	42.93±1.36	1.13±0.11a	0.46±0.02b	1.71±0.15b	0.63±0.04a
FG	<i>Stipa krylovii</i> , <i>Heteropappus altaicus</i> , <i>Artemisia frigida</i>	15.33±0.58	28.70±2.38	1.06±0.09b	0.66±0.03a	1.97±0.11a	0.46±0.02c
TA	<i>Stipa krylovii</i> , <i>Heteropappus altaicus</i> , <i>Cleistogenes squarrosa</i>	9.34±0.99	21.39±1.96	0.97±0.09b	0.37±0.01c	1.42±0.09c	0.54±0.05b

Different letters denote significant differences ($p < 0.05$) among different land use patterns.

were significantly higher than that of GE and TA ($p < 0.05$), and the difference in the Pielou evenness index was significant ($p < 0.05$), which showed that GE > TA > FG, which indicated that the most heterogeneous species distribution was found in the free-grazing area (FG).

Soil preferential flow staining characteristics and their spatial changes

Table 3 illustrates the parameters of the soil staining characteristics of the fenced area, free-grazing area, and tourist area under varying levels of infiltration. The spatial distribution of soil moisture infiltration depth in grasslands under the three utilization patterns is quite diverse, as shown by the images of soil moisture infiltration depth for GE, FG, and TA (Figure 3).

Table 3. Characteristics of preferential flow of vertical soil profiles under the three utilization patterns.

Infiltration amount(mm)	Site	Max D (cm)	DC (%)	UniFr (cm)	PF _{fr} (%)	LI (%)
10	GE	7.66±	12.49 ± 1.24a	4.22 ± 0.48a	15.41±	121.30±
		1.42b			7.45b	14.70b
	FG	11.72±	16.79 ± 2.88a	3.96 ± 1.24a	41.27±	161.15±
		1.67a			15.33a	14.22a
	TA	11.00±	13.44 ± 3.87a	3.28 ± 1.34a	40.59±	118.70±
		3.08a			7.08a	19.35b
25	GE	22.90±	29.87 ± 5.08a	5.26 ± 2.59b	56.58±	215.30±
		1.19a			17.96a	42.70a
	FG	19.34±	26.99 ± 4.46a	6.14 ± 2.57a	44.98±	163.00±
		2.53b			17.45ab	12.60b
	TA	19.04±	27.31 ± 3.32a	7.20 ± 1.66a	29.55±	162.25±
		2.69b			7.00b	20.76b
50	GE	28.40±	48.13 ± 5.44a	14.26 ± 3.35ab	35.92±	183.50±
		2.35a			8.68a	15.13a
	FG	23.84±	44.70 ± 7.70ab	15.58 ± 3.21a	23.33±	151.20±
		3.77b			2.34b	21.05b
	TA	21.36±	36.03 ± 2.89b	10.24 ± 2.82b	23.28±	159.75±
		1.37b			5.35b	13.55ab

MaxD: Maximum staining depth; DC: Dyeing area ratio; UniFr: Depth of matrix flow; PF_{fr}: Priority flow fraction; LI: length index Different letters denote significant differences ($p < 0.05$) among different land use patterns.

(a), (b), and (c) Correspond to the grassland areas under enclosure (GE), free grazing (FG), and tourism (TA) conditions, respectively, at a penetration depth of 10 mm. (a), (b) and (c) illustrate the soil water infiltration depths of the grassland in the Enclosure (GE), Free Grazing (FG) and Tourism (TA) zones, respectively, at 10 mm infiltration. (d), (e) and (f) illustrate the soil water infiltration depths of the grassland in the Enclosure (GE), Free Grazing (FG) and Tourism (TA) zones, respectively, at 25 mm infiltration. (g), (h) and (i) illustrate the soil water infiltration depths of grassland in the Enclosure (GE), Free Grazing (FG) and Tourism (TA) zones, at 50 mm infiltration, respectively.

At an infiltration volume of 10 mm, there was no significant variation in DC across the three plots (Table 3). MaxD and PF_{fr} were significantly higher for FG and TA than for GE ($p < 0.05$), while FG's LI values were significantly higher than those of GE and TA ($p < 0.05$).

Preferentially flow paths preferentially penetrated into deeper soil layers, which was consistent with the soil infiltration 3D images (Figure 3a-c). The soil matrix flow depths in the three plots did not differ significantly, indicating that the development of soil matrix flow was similar across all plots. In all three study sites, the PF_{fr} was less than 50%, indicating that the infiltration of soil moisture was primarily driven by matrix flow.

At an infiltration volume of 25 mm, MaxD increased significantly in all sample sites compared to the 10 mm infiltration volume, and the overall infiltration level of the soil increased significantly (Figures. 3d-f). The MaxD and LI values at the FG and TA sites were significantly lower than those at the GE site ($p < 0.05$), indicating that the development of preferential flow of GE was superior to that of FG and TA. The UniFr of each sample site showed TA > FG > GE, with TA and FG differing significantly from GE ($p < 0.05$). The results demonstrated a notable delay in the initiation of preferential flow in TA and FG compared to GE. The mean PF_{fr} of GE, FG, and TA were 56.58%, 44.98%, and 29.55%, respectively (Table 3). While TA was still dominated by the substrate, the PF_{fr} data showed that the preferred flow role of GE soils had surpassed that of the substrate (PF_{fr} > 50%), and that of FG was nearly equal to that of the substrate. In comparison to those under 10 mm infiltration, the DC of GE, FG, and TA were substantially higher at 29.87%, 26.99%, and 27.31%, respectively.

At an infiltration volume of 50 mm, Max D in all sample plots increased significantly compared to the 10 mm and 25 mm infiltration volumes (Figure 3g-i). The maximum infiltration depth at GE reached 31.80 cm, which was the maximum infiltration depth of the vertical profile observed in this study, and multiple vertically oriented preferential paths were observed in GE. MaxD and PF_{fr} were significantly higher in GE than in FG and TA, and no significant differences in these parameters were found between FG and TA. DC was significantly higher in GE than in TA, and no significant differences were observed among FG, GE, and TA (Table 3).

Under varying levels of infiltration water, the three study locations' variations in the soil profile staining area ratio varied significantly (Figure 4, Table 4). The ratio of the soil vertical profile-stained area increased in tandem with the amount of infiltration water. Although the vertical profile stained area ratio (DC) showed a decreasing tendency with increasing soil depth, the characteristics of the stained area ratio curves varied. At 10 mm infiltration level, both GE and TA were dominated by matrix flow, and FG began to show uneven infiltration with preferential flow in the soil layer below 4 cm. The staining depth in the soil profile progressively increased with higher infiltration quantities, as seen by the greater variation in the soil staining area ratio of GE, FG, and TA grasslands under the 25 mm and 50 mm infiltration volumes compared to the 10 mm infiltration volume. Overall, notable differences

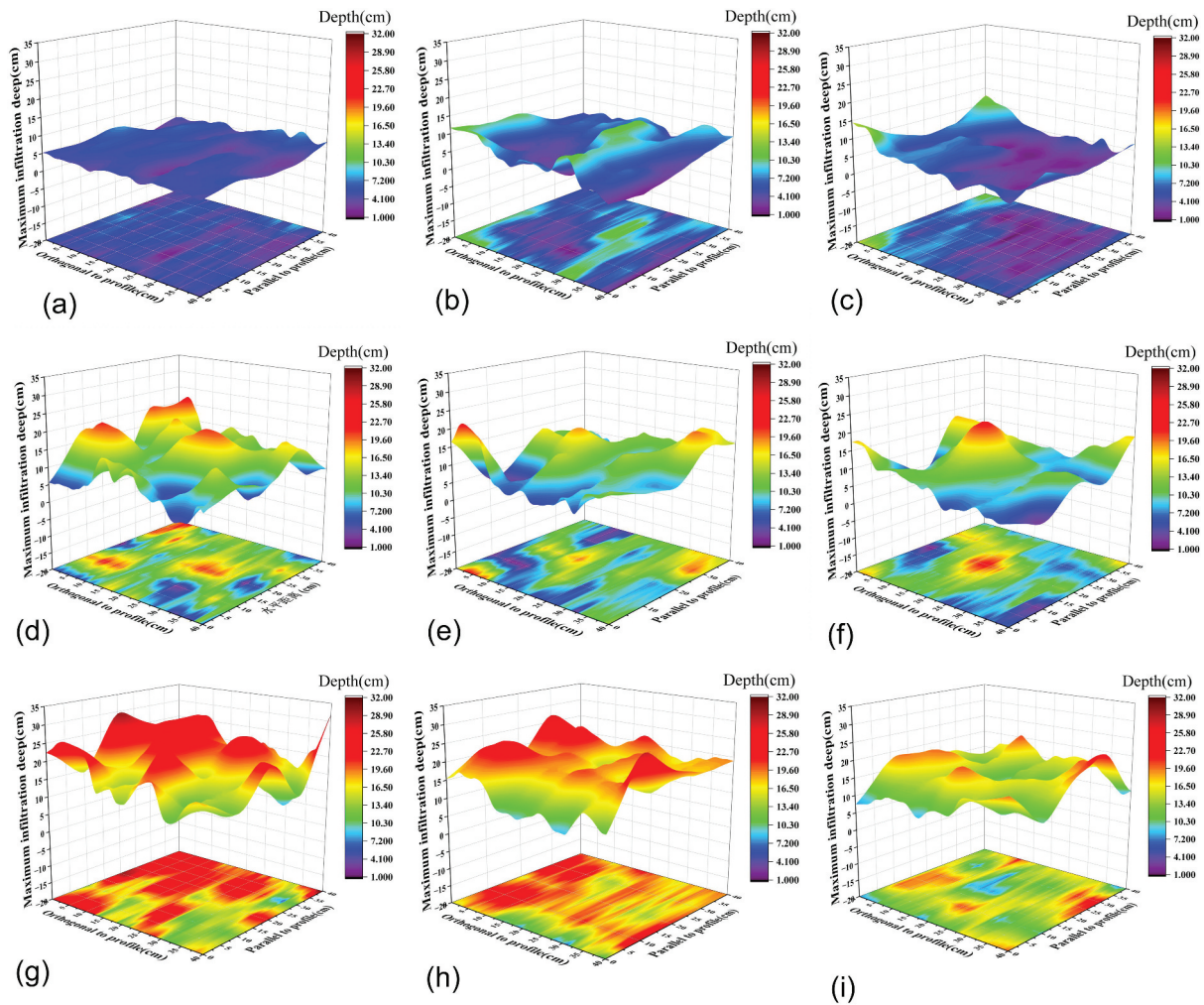


Figure 3. Characteristics of soil preferential flow spatial distribution of preferential infiltration of soil in grazing exclusion area, free grazing area, and tourist area under different infiltration volumes. (a), (b), and (c) Correspond to the grassland areas under enclosure (GE), free grazing (FG), and tourism (TA) conditions, respectively, at a penetration depth of 10 mm. (a), (b) and (c) illustrate the soil water infiltration depths of the grassland in the Enclosure (GE), Free Grazing (FG) and Tourism (TA) zones, respectively, at 10 mm infiltration. (d), (e) and (f) illustrate the soil water infiltration depths of the grassland in the Enclosure (GE), Free Grazing (FG) and Tourism (TA) zones, respectively, at 25 mm infiltration. (g), (h) and (i) illustrate the soil water infiltration depths of grassland in the Enclosure (GE), Free Grazing (FG) and Tourism (TA) zones, at 50 mm infiltration, respectively.

were observed in the water flow routes across the various soil layers, and the highest variability in the stained area ratio was observed in the grassland soil profile in the GE area. Rarely did preferential flow occur below the 30 cm depth in the research region; it mostly happened within the 0 to 30 cm soil layer.

Factors influencing soil preferential flow

The key factors influencing soil preferential flow are the continuity of preferential pathways, soil properties, and moisture content (Nimmo 2020). Previous studies have examined the effects of infiltration water on preferential flow in grassland under various usage modes. In this study, the Spearman method was employed to analyze correlations among nine factors: soil bulk density, total soil porosity, sand, silt, and clay content, saturated hydraulic conductivity, root length density, root surface area density, and root weight density, with the stained area ratio of the respective soil layer. As shown in Figure 5, a significant negative correlation ($p < 0.05$) was found between the stained area ratio and soil bulk density; higher bulk density correlated with lower soil porosity, which hinders preferential flow. In sandy soils, increased clay content helps create better

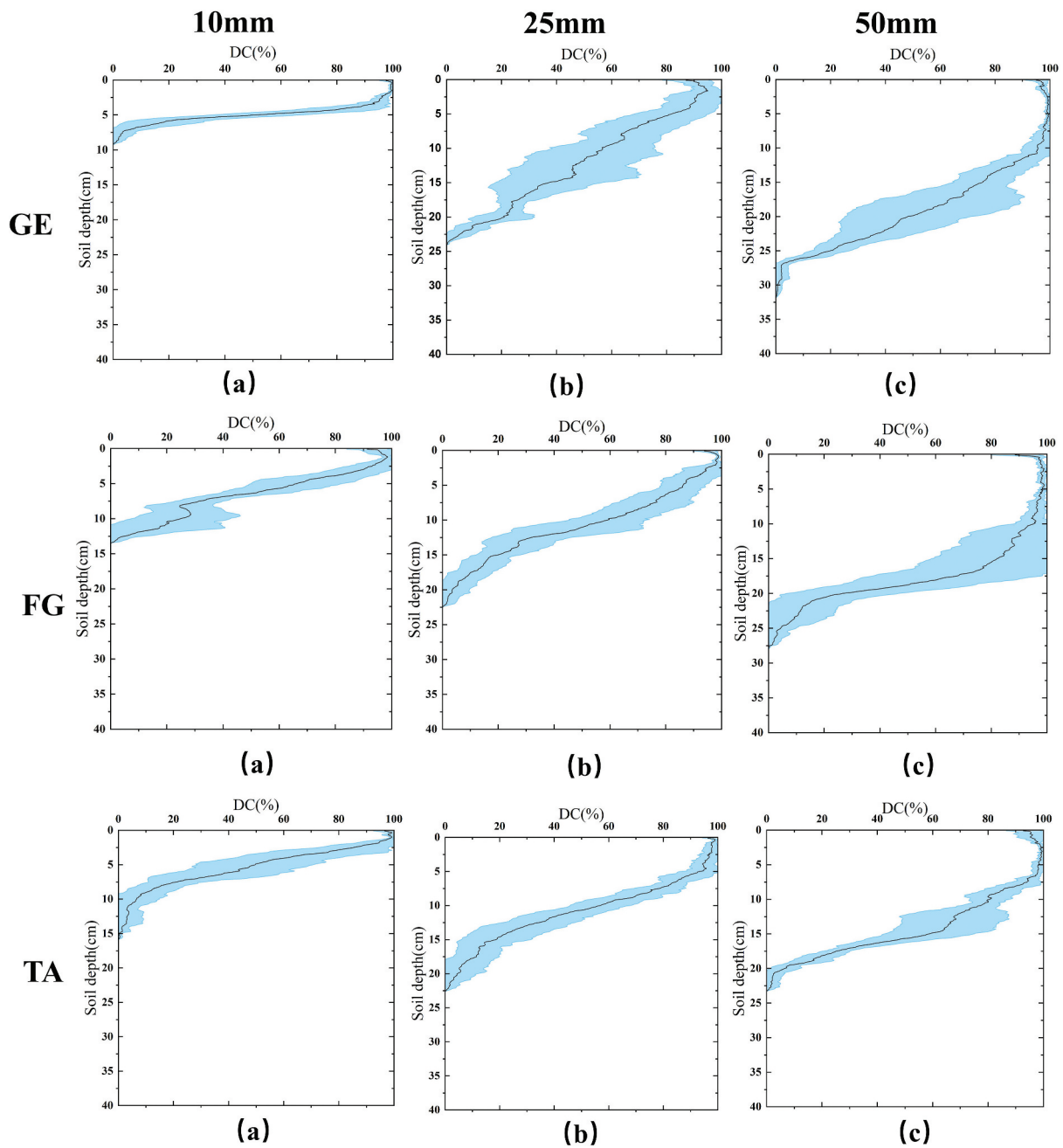


Figure 4. Variation of soil profile dyeing area ratio with soil layer under different infiltration conditions. Stained area ratios in the figure are mean $\pm$ standard deviation (blue areas). a, b, and c are stained area ratios of soil profiles after 10 mm, 25 mm, and 50 mm infiltration water treatments, respectively.

Table 4. Dyeing area ratio in grazing exclusion area, free grazing area, and tourist area under different infiltration volumes.

site	Infiltration amount(mm)	DC (%)
GE	10	12.49 $\pm$ 1.24a
	20	29.87 $\pm$ 5.08b
	30	48.13 $\pm$ 5.44b
FG	10	16.79 $\pm$ 2.88a
	20	26.99 $\pm$ 4.46b
	30	44.70 $\pm$ 7.70c
TA	10	13.44 $\pm$ 3.87a
	20	27.31 $\pm$ 3.32b
	30	36.03 $\pm$ 2.89c

Dyeing area ratio (DC). Different letters denote significant differences ($p < 0.05$) among different land use patterns.

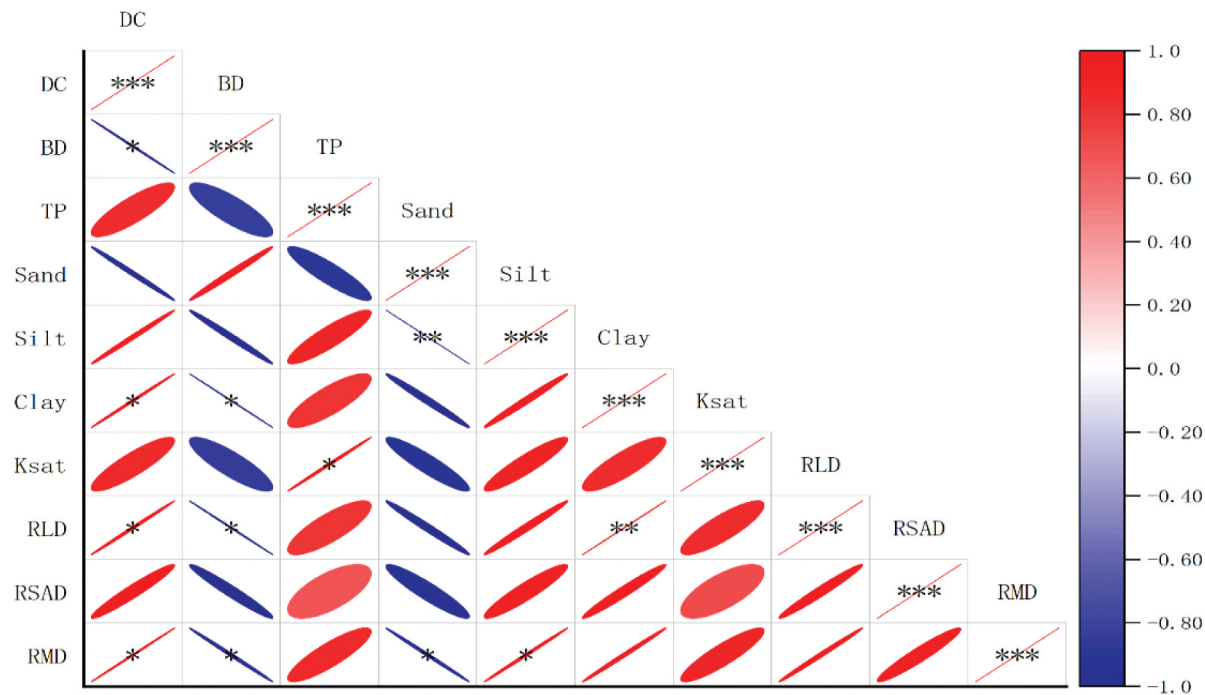


Figure 5. Spearman correlation analysis of soil and root characteristics on dyeing area ratio. Significant correlation is shown at the 0.05 level by the symbol *, at the 0.01 level by the symbol **, and at the 0.001 level by the symbol ***. DC denotes Dye Area Ratio, BD denotes Soil Bulk Density, TP denotes Total Soil Porosity, Sand denotes Sand Fraction, Silt denotes Silt Fraction, Clay denotes Clay Fraction, Ksat denotes Saturated Hydraulic Conductivity, RLD denotes Root Length Density, RSAD denotes Root Surface Area Density, RMD denotes Root Mass Density.

pore structures, facilitating preferential flow. Additionally, a significant positive correlation ($p < 0.05$) was observed between the stained area ratio and both root weight density and root length density. The results of the study revealed that the primary factors affecting preferential flow in semi-arid grassland are soil bulk density, clay content, root length density, and root weight density.

Discussion

Soil and vegetation characteristics of semi-arid grasslands under different utilization

In semi-arid grassland ecosystems, soil physical properties are influenced by various factors, including grazing intensity, litter cover, root system, geoclimatic conditions, and human activities (Jiang et al. 2023). Among these, land use practices play a significant role in altering soil properties (Sun et al. 2018). It was found that different land use practices significantly affected soil properties across the GE, CG, and FG plots. Implementing fencing and sealing measures resulted in improved physical properties of the topsoil. Notably, soil bulk density was significantly lower in the GE compared to the FG and TA plots, while total soil porosity showed a marked increase. However, livestock trampling and tourism activities led to soil compaction and deformation, which increased bulk density and disrupted pore connectivity (Zhao et al. 2010). These findings suggest that the soil health deteriorates due to grazing and tourism, which aligns with previous research conducted on Inner Mongolian grasslands (Abdalla et al. 2018).

Different utilization modes significantly influence grassland species composition and diversity. This study revealed that the plant height and vegetation cover in the GE were significantly greater than in the FG and TA, aligning with previous research on Inner Mongolia grasslands (Du and Gao 2021). After implementing fencing and sealing measures, the vegetation cover, height, and aboveground biomass in severely damaged grasslands showed significant improvement, alongside enhanced soil physical properties (Liu et al. 2019). However, opinions differ regarding the effects of these measures

on plant community diversity and richness. While many studies indicate that fencing can enhance vegetation diversity and richness, others suggest that fenced areas may exhibit lower diversity compared to grazed or tourist-affected grasslands, resulting in a more homogeneous community structure (Yang et al. 2017; Li et al. 2023). Consistent with the findings of Du et al., this study revealed that the Shannon-Wiener and Simpson indices for FG were significantly higher than those for GE ($p < 0.05$) (Du and Gao 2021). Field surveys indicated that dominant plant species in the enclosure, such as Kirschner's needlegrass and goatgrass, were more competitive for essential resources like water and heat. This competitive advantage hindered the growth of less competitive species, leading to a decline in overall vegetation diversity.

During the staining tracer test and subsequent image analysis, it was found that the root systems of semi-arid grassland serve as crucial pathways for preferential flow. The root surface area density across different diameter classes consistently decreased with increasing soil depth at each sample site, a trend that aligns with findings by Xu et al. (2022). Additionally, at a depth of 0–10 cm, root diameter, root length density (RLD), and root mass density (RMD) were significantly higher in the GE compared to the FG and TA. However, this difference diminished at depths of 10–30 cm. This trend likely results from enclosure measures that greatly enhance the physicochemical properties and moisture content of the upper soil layer, while deeper layers show minimal changes (Chen et al. 2022). Other researchers have also arrived at similar conclusions (Chen and Tang 2016).

Effects of different utilization modes on the staining characteristics of grassland soil preferential flow

The development of preferential flow in grasslands varies significantly among different utilization modes. With 10 mm of infiltration water, the UniFr and MaxD of grassland showed minimal differences under three land utilization modes. However, a notable variation was observed in the degree of preferential flow differentiation, ranked as $FG > TA > GE$. This finding contrasts with previous research, such as that by Wu et al., which suggests that preferential flow is generally more pronounced in fenced areas compared to free-grazing zones (Wu et al. 2022). Additionally, the results indicated that water movement within the soil profile remained predominantly governed by matrix flow. This may be attributed to the limited infiltration depth caused by the small volume of water, where preferential flow is primarily influenced by the structure of the surface soil (Nimmo 2020). In the GE, higher vegetation cover contributes to a loose, pore-rich surface soil, allowing more water to infiltrate uniformly into the soil matrix when infiltration is minimal. As a result, the development of preferential flow in GE is lower compared to FG and TA (Budhathoki et al. 2022).

At higher infiltration levels, rainfall can quickly saturate the top soil layer and excess water begins to flow into the macropores in the soil (Zhang et al. 2022). At 25 mm infiltration volume, the average infiltration depth and the extent of preferential flow development of grassland increased significantly compared with those at 10 mm infiltration volume, and the soil profile in the GE exhibited a predominance of preferential flow. The depth of preferential flow in the soil profile exceeds that of the matrix layer, and the preferential flow rate was about three to four times that of the substrate flow rate. Preferential flow in the lower soil layer was encouraged by the rise in infiltration water, which also enhanced the development of matrix flow (Kulli et al. 2003).

Many studies suggest that preferential flow is more prevalent during periods of high rainfall (2020; Zhang et al. 2022). However, this study revealed that at an infiltration volume of 50 mm, both the overall depth of infiltration and the depth of substrate flow in the soil profile significantly increased, while the proportion of preferential flow decreased compared to the 25 mm infiltration volume. This finding indicates that the increase in water infiltration did not significantly enhance the capacity for preferential water transport, suggesting that infiltration volume is no longer the primary limiting factor for preferential flow development (Zhang et al. 2018). One possible explanation for this result is that herbaceous plants typically have shorter root lengths (Kan et al. 2019). At an infiltration volume of 50 mm, although the overall depth of soil moisture infiltration increased, the uniformity of infiltration decreased due to the absence of preferential pathways, such as root pores, in the deeper soil layers. In this study, it was found that preferential flow primarily occurred within the 10–25 cm range of the vertical profile, which aligns closely with the root growth range of semi-arid grassland herbaceous plants. This suggests that the development of preferential flow is significantly influenced by the plant root system, a finding that corroborates previous research (Wu et al. 2022).

At both 25 mm and 50 mm infiltration volumes, the MaxD, DC, Li, and PF-fr in the GE were significantly greater than those in the FG and TA. This discrepancy can be attributed to the high vegetation cover in the enclosed area, along with an extensive underground root system that creates large pores conducive to preferential flow (Wang et al. 2023). In contrast, free grazing areas and tourist areas tend to hinder the formation of a macroporous structure due to compacting by animals and tourists (Greenwood and McKenzie 2001; Zhao et al. 2010). Singleton noted that in continuously grazed regions, the porosity of the topsoil and vertical pore continuity are adversely affected by the breakdown of larger soil aggregates, which are then replaced by smaller aggregates that fill existing pores (Singleton et al. 2000). This process significantly influences the development of preferential flow pathways (Singleton et al. 2000). In summary, both soil preferential flow and matrix flow occurred simultaneously during the infiltration process in the study area. Under identical external water supply conditions, the degree of preferential flow development in arid grasslands varies by land use type. Specifically, at infiltration rates of 25 mm and 50 mm, enclosed areas exhibit the highest differentiation in preferential flow routes, while free-grazing and tourist areas show less pronounced differentiation. Dye tracer patterns reveal that matrix flow primarily occurs in the surface soil layer, whereas preferential flow is more prevalent at depths of 10 to 30 cm. However, during periods of low rainfall, the advantage of preferential flow development in enclosed areas is not significant.

Moreover, plant root channels serve as the principal pathways for preferential water movement (Li et al. 2020). In arid grasslands, an increase in infiltration water does not always lead to a substantial enhancement in preferential water movement, as this is often limited by root length. Consistent with prior studies, this research highlights that the effects of vegetation cover, land use patterns, and precipitation levels on preferential flow can vary significantly across different regions (Yan and Zhao 2016; Zhang et al. 2018). These findings indicate that the influence of each factor on preferential flow is unique and context-dependent based on the specific characteristics of the study area.

Impact of soil and vegetation characteristics on prioritized behaviors

Grazing and tourism significantly alter the plant community characteristics in grassland ecosystems, leading to declines in species composition, vegetation cover, and height. These changes in plant communities subsequently impact the structural properties of grassland soils (Du and Gao 2021). Variations in soil texture and structure result in differing water-holding capacities and internal pore structures, which in turn affect the formation and development of preferential flow (Chen and Tang 2016). Previous studies have shown that soils with high sand content exhibit rapid infiltration rates. An increase in the proportion of sand particles and a decrease in the proportion of silt particles can enhance the likelihood of preferential flow occurrence (Li et al. 2020). However, some researchers have also suggested that soils with high clay content tend to develop distinct layered structures, which can promote the formation of preferential flow (Zhao et al. 2022). This study corroborates the positive correlation between soil clay content and the staining area ratio. As the clay particle content in the soil increases, more stable soil aggregates are formed, which facilitates the creation of larger pores, thereby promoting the development of preferential flow (Radolinski et al. 2022). Additionally, it has been established that root activities during plant growth contribute to the formation of large, low-tortuosity, and highly interconnected pores in the soil, which are crucial for driving the preferential flow of soil moisture (Clothier et al. 2007; Guo et al. 2019). However, there remains a gap in research specifically examining which root traits significantly affect soil preferential flow.

This study highlights the significant role of semi-arid grassland plant root systems in influencing the development of preferential flow. Both RLD and RMD exhibit a positive correlation with the stained area ratio. A higher RMD indicates a well-developed underground root system, resulting in a greater number of channels created by root growth. Similarly, increased RLD suggests improved connectivity among these channels, which facilitates preferential flow (Nimmo 2020). In the GE, the root systems demonstrate greater density and length compared to those in free-grazing and tourist areas, leading to the formation of larger voids and pore sizes. This condition enhances the likelihood of preferential flow occurring within the soil (Zhang et al. 2018). Additionally, the GE facilitates preferential flow development more effectively than FG and TA, resulting in improved soil water infiltration and increased water availability for vegetation (Li et al. 2020). This enhanced water availability fosters greater productivity

and accelerates nutrient cycling. Such interactions may create a feedback loop that supports vegetation growth during periods of resource scarcity (Valani et al. 2022).

Limitations and future research

This study acknowledges several limitations related to experimental conditions, equipment, and the number of researchers involved, which should be addressed in future research. (1) The dye tracer method was utilized to illustrate the final morphology of dye distribution within the sample plot via vertical profiles. However, this method has limitations in accurately simulating the dynamic changes associated with preferential flow in grassland soils and the mechanisms of surface runoff during actual rainfall events. To enhance understanding, future studies should incorporate artificial rainfall experiments to investigate preferential flow more comprehensively, along with real-time monitoring of moisture content variations during the dyeing process. This approach could provide deeper insights into soil preferential flow dynamics in the study area. (2) While this study explored preferential flow characteristics under various land use patterns, offering valuable theoretical support for grassland management and regional water resource protection, the relatively short time frame may have overlooked the effects of different plant growth seasons on soil preferential flow dynamics. Additionally, given the limited experimental period, future research should aim for long-term, larger-scale field surveys to gather more representative data.

Conclusions

Activities like grazing and tourism significantly alter the characteristics of plant communities in grassland ecosystems. These disturbances result in decreased vegetation cover and accelerated soil degradation, which in turn affect regional ecohydrological processes. The development of preferential flow in grasslands varies significantly depending on land use practices. At infiltration levels of 25 mm and 50 mm, the extent of preferential flow differed across various usage types. Enclosed grasslands showed the highest preferential flow at both infiltration rates, followed by grazed areas, while tourism-affected regions exhibited the least. Increasing the volume of infiltration enhances the progression of matrix flow and encourages the emergence of preferential flow in deeper soil layers. With a 10 mm infiltration volume, water primarily seeps into the surface soil, leading to high retention and minimal preferential flow development. In contrast, at 25 mm and 50 mm infiltration levels, a marked increase in areas displaying preferential flow is observed within the vertical soil profile. As infiltration volume rises, the maximum depth of water infiltration also increases; however, the differentiation in preferential flow does not consistently align with the changes in infiltration volume. Contrary to conventional understanding, we found that increased infiltration volume does not consistently enhance preferential flow development, suggesting the presence of complex threshold behaviors in soil moisture movement.

In semi-arid grassland soils, various factors influence preferential flow, including vegetation type, root distribution, and soil properties. Notably, soil bulk density, clay content, root length density, and root weight density exhibit significant correlations with the occurrence of preferential flow. Furthermore, greater vegetation cover supports the development of preferential flow within the soil.

Based on the findings of this study, several practical implications can be derived to guide sustainable grassland management and water resource conservation in semi-arid regions. Enclosed grasslands, compared to free grazing and tourism, significantly improve water infiltration and preferential flow. This enhancement is crucial for replenishing deep soil moisture and aiding vegetation recovery in semi-arid areas. We recommend that relevant authorities adopt enclosure strategies for the natural restoration of degraded grasslands. Additionally, sustainable development policies, such as ecotourism and rotational grazing, should be explored to enhance the economic livelihoods of farmers and herders while protecting and restoring grassland ecosystems. These insights provide practical guidance for policymakers and land managers in designing targeted conservation strategies, promoting ecotourism, and balancing ecological restoration with socioeconomic development in water-limited grassland ecosystems. Future research should incorporate multi-scale approaches (from micro to macro) and develop innovative real-time monitoring technologies to improve the understanding and prediction of preferential flow dynamics.

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Methodology: Yan Liu

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All authors have read and agreed to the published version of the manuscript.

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Data availability statement

The dataset supporting the findings of this study is publicly available at Zenodo under the DOI:

<https://doi.org/10.5281/zenodo.13968271>. The data can be accessed and used under the terms provided by the repository.

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